Bachelor thesis plan

Spring 2026 - Plan submission deadline: 16/01/2026
Department of Advanced Computing Sciences
Board of Examiners

Personal Details

Name: *(Student's family name and first names)*
Jake Oliver Lockitch

Student number: *(Student's ID number as provided by university)*
6405499

Start date of thesis research: *(Cannot be prior to receiving BoE permission, hence must be in the future)*

Expected graduation date: *(Date of the regular bachelor conference where the thesis will be defended:*

- *Regular defence: 5-17 June 2026 (TBA) / Resit defence: 24-28 August 2026 (TBA)*

5-17 June 2026

Programme: please select from drop-down list

☐ The student is enrolled in the KE@Work/CS@Work programme *(check if applicable)*
☐ The student is enrolled in the Marble 2.0 programme *(check if applicable)*

Supervision details

The thesis supervisors are DACS staff members that give process based and content-based supervision. At least one of the thesis supervisor acts as an examiner. External supervisors are listed in the next section.

Name DACS thesis supervisor 1: Mirela Popa
Name DACS thesis supervisor 2: Petra Bosilj
Name DACS thesis supervisor 3: optional

Frequency of contact between DACS thesis supervisor(s) and student: multiple times a week

External collaboration

☐ This thesis involves an external collaboration *(check if an external party if actively involved)*

In case this thesis is not part of an external collaboration proceed to the section "assessment".

Name external institute: *(Name the institute of external collaboration. In the case of multiple institutes, name the most important one and add an appendix with an overview of external parties)*

This is: please select from drop-down list

Location external institute:

Name external supervisor:

Frequency of contact between the external supervisor(s)/advisor(s) and student:

Frequency of contact between the external supervisor(s)/advisor(s) and DACS thesis advisor(s):

☐ The student will be hosted by/have a secondment at the external institute *(Check this if the student will work as an internal at the external institute or be visiting the external institute for a period larger than two weeks)*

The student will be hosted by the external institute for:

☐ The entire period

☐ A period of weeks

During this period, the work times of the student are as follows:

☐ There is a confidentiality agreement with the following clauses: *(check when applicable)*

☐ The code developed will remain property of the external institute

☐ Special permission has been obtained for limited confidentiality *(See Rules and Regulations and attach documents)*

Note that all thesis must be sent to the DACS Student Affairs Office for storing, where they can be labelled "confidential", but DACS committees and assessment committees (visitatie commissies) will always have (confidential) access for auditing purposes.

UM emphasizes that it is in your best interest to have the standardised UM threefold internship agreement signed, so that your rights and duties for an internship are ensured in accordance with UM standards. You can find this agreement (and more important information) here: sign in to the Maastricht University intranet and navigate to Home > DACS students > My studies > Thesis bachelor and on the page, there is a section which includes the form. Please select the agreement corresponding to your internship location, whether it is <u>within</u> or <u>outside</u> the Netherlands.

On behalf of the external institute, , consents with this thesis plan

Signature:

Date of signing:

Assessment

The assessment will be based on:

(Specify the terms of assessment that at least should contain the thesis, process and the presentation)

☒ The contents and form of the thesis, which must at least be sufficient for passing the thesis

☒ A (software) product that accompanies the thesis

☒ The presentation

☒ The process

☐ (optional)

Thesis details

Thesis title: *(Working title of the thesis)*

Improving Robustness And Practical Viability Of Ai-Based Cctv Violence/Incident Detection Under Township-Like Conditions In South Africa

Description of the assignment: *(A single page, describing the assignment and indications of the techniques that will be explored)*

This thesis evaluates whether AI-based CCTV violence, crime and aggression detection can be technically and operationally feasible in South African townships. In practice CCTV footage in townships involves low resolution, low light, blocked crowd and objects and has to deal with frequent power outages (load-shedding). These factors directly cause a "transferability" issue: a model may perform very well with controlled training, but results can degrade hugely under real-world conditions.

Therefore, we will go further than just applying pre-trained models in a township. The goal is to create technical novelty by coming up with and evaluating an adapted architecture and training approach to improve robustness and accuracy in township contexts. The work will be conducted using public CCTV-style datasets tagged with violence vs non-violence / incident vs non-incident. This enables reproducible evaluation without requiring sensitive local CCTV data.

This thesis will consist of four components:

Mapping existing approaches and evidence quality: Conduct a structured review of AI-enabled CCTV violence/incident detection research and reported deployments. Extract model families, data assumptions and metrics used. Distinguish benchmark-only evidence from field evaluations.

Baseline implementation and evaluation on public annotated datasets: Reproduce at least one representative baseline model for video violence/incident detection. Establish a transparent evaluation pipeline and report baseline results.

Township robustness benchmark via controlled degradations: Create a township-like stress test benchmark by systematically degrading the videos (down sampling, compression, low-light, blur etc). Quantify the decrease in performance and analyse failure models focusing on false positives. This benchmark will be referred to as the Township Corruption Benchmark and is a core contribution of this thesis, inspired by video corruption benchmarks. Chang et al. (2020) study realistic video file corruptions and report large accuracy drops in action recognition, motivating corruption-oriented evaluation beyond clean benchmarks.

Technically novel robustness improvement and viability assessment: Propose and evaluate an adapted architecture and training method aimed at improving robustness under degradations. Compare results against baselines and report improvements. Robustness training will be included by training with both clean and degraded data. Similar to corruption-aware augmentation strategies like PRIME (Modas et al., 2022).

Datasets (public, annotated and CCTV-style):

Bus Violence (Ciampi et al., 2022): 1400 videos (700 violence/700 non-violence) recorded from three cameras in a bus. Variation of lighting and backgrounds.

Real-Life Violence Situations (RLVS): 2000 videos collected from real world scenes.

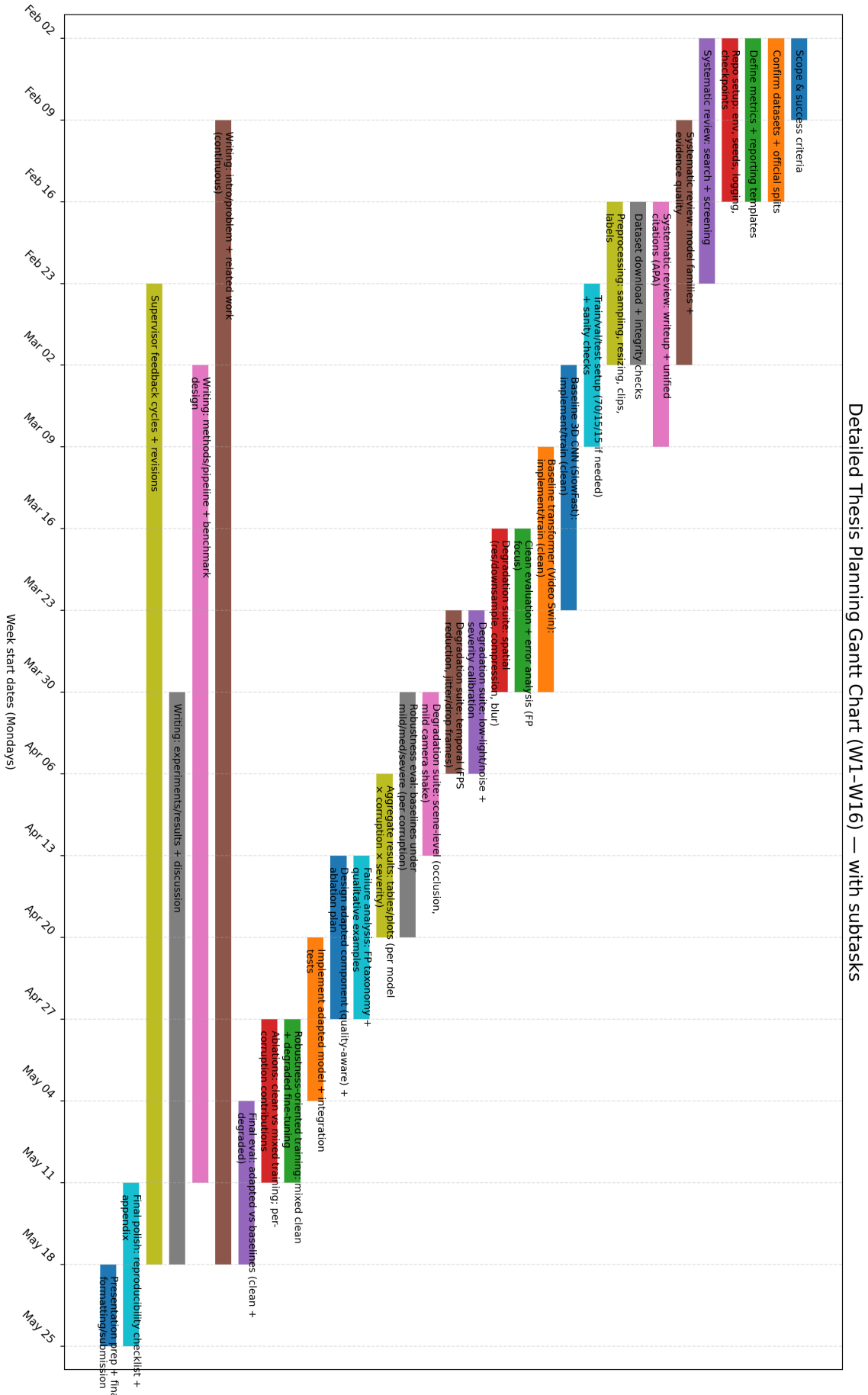
Surveillance Camera Fight dataset: 300 videos (150 fight / 150 non-fight). Surveillance footage.

Baseline models/architecture: I3D (inflated 3D ConvNet), SlowFast Networks. 3D CNNs are strong baselines on violence datasets and have been used in Bus Violence experiments (Ciampi et al., 2022). Along with a 3D CNN baseline, a modern transformer baseline will be included to compare robustness behaviour under corruptions. For the baseline comparison, I will pick only one model for each family: SlowFast Networks as the 3D CNN baseline (Feichtenhofer et al., 2019) and Video Swin Transformer as the transformer baseline (Liu et al., 2022).

Experiments to answer research questions:

Standard evaluation: training/validate/test on each dataset using official splits when available or use (70/15/15) with a random seed then report baseline performance.
Township-like robustness benchmark: edit the video quality down sample / compress / reduce brightness / motion blur / occlusions / reduce FPS, on 3 levels: mild, medium and severe. Township corruption taxonomy examples: spatial quality: resolution down sampling (e.g. 224→112→64). JPEG/MPEG compression (different bitrates) / motion blur / low-light / reduced FPS (simulate cheap cameras and dropped frames) and scene-level: synthetic occlusion (random rectangles, blocking view), mild camera shake.
Technical novelty: propose and evaluate an adapted system and training approach to improve robustness under township like scenarios, for example adding a quality-aware component to deal with low video quality. Compare the adapted system against the baseline models on both clean and degraded videos and check if robustness improved without an unacceptable increase in false positives.
Evaluation: report accuracy, precision, recall and F1 on each dataset. Also ROC/AUC and confusion matrices looking specifically at false positives. Evaluate clean vs degraded videos (mild, medium and severe) for each corruption type, and summarise results in a table showing performance of each model under each condition so an answer can be given about the extent to which this problem can be solved under various conditions.
References (APA):
Carreira, J., & Zisserman, A. (2017). Quo Vadis, Action Recognition? A New Model and the Kinetics Dataset. Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR).
Chang, T., Fu, D. Y., Li, Y., & Ré, C. (2020). Beyond the Pixels: Exploring the Effect of Video File Corruptions on Model Robustness. ECCV Workshop on Adversarial Robustness in the Real World (AROW).
Ciampi, L., Foszner, P., Messina, N., Staniszewski, M., Gennaro, C., Falchi, F., Serao, G., Cogiel, M., Golba, D., Szczęsna, A., & Amato, G. (2022). Bus Violence: An Open Benchmark for Video Violence Detection on Public Transport. Sensors, 22(21), 8345. https://doi.org/10.3390/s22218345
Feichtenhofer, C., Fan, H., Malik, J., & He, K. (2019). SlowFast Networks for Video Recognition. Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV).
Liu, Z., Ning, J., Cao, Y., Wei, Y., Zhang, Z., Lin, S., & Hu, H. (2022). Video Swin Transformer. Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR).
Modas, A., Rade, R., Ortiz-Jiménez, G., Moosavi-Dezfooli, S.-M., & Frossard, P. (2022). PRIME: A Few Primitives Can Boost Robustness to Common Corruptions. European Conference on Computer Vision (ECCV).

Planning: *(Provide a complete planning for the thesis period with a reasonable time path for completion, including time for writing the thesis. The preferred format is a Gantt chart that you can attach separately)*


Detailed Thesis Planning Gantt Chart (W1–W16) — with subtasks
Scope & success criteria
Confirm datasets + official splits
Define metrics + reporting templates
Repo setup: env, seeds, logging, checkpoints
Systematic review: search + screening
Systematic review: model families + evidence quality
Systematic review: writeup + unified citations (APA)
Dataset download + integrity checks
Preprocessing: sampling, resizing, clips, labels
Train/val/test setup (70/15/15 if needed) + sanity checks
Baseline 3D CNN (SlowFast): implement/train (clean)
Baseline transformer (Video Swin): implement/train (clean)
Clean evaluation + error analysis (FP focus)
Degradation suite: spatial (res/downsample, compression, blur)
Degradation suite: low-light/noise + severity calibration
Degradation suite: temporal (FPS reduction, jitter/drop frames)
Degradation suite: scene-level (occlusion, mild camera shake)
Robustness eval: baselines under mild/med/severe (per corruption)
Aggregate results: tables/plots (per model x corruption x severity)
Failure analysis: FP taxonomy + qualitative examples
Design adapted component (quality-aware) + ablation plan
Implement adapted model + integration tests
Robustness-oriented training: mixed clean + degraded fine-tuning
Ablations: clean vs mixed training; per-corruption contributions
Final eval: adapted vs baselines (clean + degraded)
Final polish: reproducibility checklist + appendix
Presentation prep + final formatting/submission
Writing: intro/problem + related work (continuous)
Writing: methods/pipeline + benchmark design
Writing: experiments/results + discussion
Supervisor feedback cycles + revisions
Feb 02
Feb 09
Feb 16
Feb 23
Mar 02
Mar 09
Mar 16
Mar 23
Mar 30
Apr 06
Apr 13
Apr 20
Apr 27
May 04
May 11
May 18
May 25
Week start dates (Mondays)

Problem statement: *(Single sentence describing the overall challenge faced by the student)*
AI-based CCTV violence/incident detection often shows high benchmark performance, but its reliability and performance decreases under low-quality, crowded, and resource-constrained conditions, making it unclear whether it can be viable in township-like environments without targeted robustness improvements.

Research questions: *(Provide two to five research questions of a single sentence that by answering them, jointly provide a path for addressing the problem statement)*

1. How do representative baseline CCTV violence/incident detection models perform on public annotated CCTV-style datasets under standard evaluation, and what patterns of false positives and false negatives emerge?
2. To what extent does performance degrade when the same models are evaluated under township-like video conditions (e.g., low resolution, low-light, compression, blur, occlusion, reduced FPS), and which failure modes are most common at different severity levels?
3. Can an adapted architecture or robustness-oriented training strategy improve accuracy and stability under degraded conditions compared to baselines, without causing unacceptable increases in false alarms, computational cost, or inference latency?
4. Which specific degradation types contribute most to real-world transferability failures, and which mitigation strategies provide the best trade-off between performance on clean footage and robustness against degraded township-like conditions (especially in terms of false positives vs. missed incidents)?
5. (optional)?

Alignment with Programme Objectives and Learning Outcomes: *(Please specify in a few lines how the thesis topic relates to the objectives and courses in your programme)*
This thesis aligns with Computer Science as it will be incorperating AI and machine learning along with computer vision into a real world safety related problem. It includes research design, reproducing software development, robustness analysis. So it overall includes knowledge from machine learning, computer vision, data analysis and it will all be written in a scientific report which will follow acedemic standards.

Signature

The student would like to make the following additional statements to the Board of Examiners regarding this thesis plan: (optional)

Ensure that you: *(Ensure that you check all)*

- ☒ are familiar with the Education and Examination Regulations (EER) that are published on student portal, as is expected of each student
- ☒ are familiar with the Rules and Regulations that are published on student portal, as is expected of each student
- ☒ are familiar with the concept of plagiarism and will refrain from committing it
- ☒ are aware of academic standards, in particular relating to proper citing
- ☒ have filled out the complete form
- ☒ have included a planning and/or attached a Gantt chart with a planning for your thesis
- ☐ have collected the signatures of both examiners and if applicable, the external institute

To avoid students starting too late with their thesis, the deadline for the thesis plans has been set by the thesis coordinator before some of the grades of the preceding semester have come in. Therefore, the possibility of getting conditional approval has been given.

In separate, motivated requests to the Board of Examiners, dispensation from this rule can be requested for cases of hardship. Such motivated requests can be submitted to dacs-boe@maastrichtuniversity.nl and are accompanied by a study plan.

☒ I understand that I must have attained 138 ECTS already for this bachelor programme, of which 60 ECTS have been obtained in year 1 and at least 40 ECTS in year 2. I declare that (check one of the following):

☐ I have already obtained the required ECTS
☒ I do not yet have the required number of ECTS, but expect that once the grades of the resits, project/KE@work/MaRBLe 2.0 that took place in the semester preceding the thesis semester come in, I will have enough ECTS. Hence, I'm submitting for conditional approval and understand that, if the expected ECTS are not obtained, any conditional approval of the thesis will be automatically withdrawn.
☐ I do not have enough ECTS; but submitting this plan together with a motivated request for dispensation due to hardship and an individual study plan.

Date: 16/01/2026

Student's signature:

Examiners

There should be two thesis examiners that approve of this thesis plan. At least one of the examiners acts as a thesis supervisor. The examiners must be DACS staff members (lecturer, assistant, associate, or full professor) with a PhD degree, a uniform teaching qualification (UTQ) and are involved in the programme. See Rules and Regulations for additional conditions. Postdoctoral researchers can only act as second examiners as part of a training programme.

The examiners have checked the form for the requirements in the EER and R&R and checked it with the ILOs. The examiners are also available in Maastricht for the bachelor conference indicated.

Name proposed thesis examiner 1:	Signature:	Date of signing:

Name proposed thesis examiner 2:	Signature:	Date of signing: